THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 1

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The present ages of Ria and Sia are in the ratio 3:5. After 8 years the ratio will be 2:3. Ria's present age (years) is: 1

(a) 40 **(b)** 24 **(c)** 32 **(d)** 22

2. ₹ 4500 is divided among three friends in the ratio 2:3:5. The difference between the largest and the smallest share (in ₹) is: 1

(a) 1350 **(b)** 1351 **(c)** 2250 **(d)** 900

3. In a pie chart a 45° sector represents 13 students. The total number of students is: 1

(a) 117 **(b)** 104 **(c)** 468 **(d)** 52

4. If $a : b = 3 : 5$ and $b : c = 10 : 7$ and $a + b + c = 322$, then b equals: 1

(a) 84 **(b)** 140 **(c)** 98 **(d)** 154

5. 120 men have provisions for 30 days. After 10 days 20 men leave. The remaining provisions last the rest of the men (days): 1

(a) 36 **(b)** 26 **(c)** 24 **(d)** 20

6. A vessel has 36 L milk and 12 L water. The litres of water to be added to make the ratio of milk to water 2:1 is: 1

(a) 12 **(b)** 9 **(c)** 6 **(d)** 24

7. The wages of 12 workers for 20 days are ₹ 36000. The wages of 9 workers for 14 days at the same daily rate (in ₹) are: 1

(a) 36000 **(b)** 19050 **(c)** 18900 **(d)** 27000

8. The present ages of Ria and Sia are in the ratio 2:5. After 9 years the ratio will be 7:13. Ria's present age (years) is: 1

(a) 10 **(b)** 21 **(c)** 30 **(d)** 12

9. **Assertion (A):** If 5 kg of rice costs ₹300, then 8 kg of rice costs ₹480. 1

Reason (R): The cost of rice is directly proportional to its quantity, so $\frac{\text{cost}}{\text{quantity}}$ is constant.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** For a fixed distance, the speed of a vehicle and the time taken are inversely proportional. 1

Reason (R): For a fixed time, the work done by workers is directly proportional to the number of workers.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. If $x : y = 3 : 4$, find the ratio $(5x + 2y) : (4x - 1y)$. 2

12. Two numbers are in the ratio $7 : 3$ and their difference is 52. Find the numbers. 2

13. The present ages of a father and son are in the ratio 7:2. After 6 years the ratio will be 8:3. Find their present ages. 2

14. If $x : y = 5 : 2$, find the ratio $(3x + 4y) : (2x - 3y)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A garrison of 240 men has provisions for 45 days. After 15 days 60 men leave. For how many more days will the remaining provisions last for the men who stay? 3

16. A mixture contains milk and water in the ratio 3:2. When 10 L of water is added, the ratio of milk to water becomes 3:4. Find the quantity of milk and water in the original mixture. 3

17. In a pie chart, the sector for Science is 40° and represents 12 students, while the sector for Maths is 80°. Find the total number of students and the number who like Maths. 3

OR

A garrison of 180 men has provisions for 40 days. After 10 days 45 men leave. For how many more days will the remaining provisions last for the men who stay? 3

18. A mixture contains milk and water in the ratio 5:3. When 12 L of water is added, the ratio of milk to water becomes 5:6. Find the quantity of milk and water in the original mixture. 3

OR

In a pie chart, the sector for Science is 55° and represents 11 students, while the sector for Maths is 135°. Find the total number of students and the number who like Maths. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** The scale of a road map of Rajasthan is 1 cm : 20 km. Two towns are 7.5 cm apart on the map. 5

(i) Find the actual distance between the towns.

(ii) Two other towns are 260 km apart. What is their distance on the map?

(iii) Write the scale as a ratio of the same units.

(iv) A bus covers the first distance at 50 km/h. How long does it take, and how long would it take at 75 km/h?

20. The present ages of Anil and Bina are in the ratio 3:7. After 5 years the ratio of their ages will be 1:2. Find their present ages and the ratio of their ages 5 years ago. 5

OR

30 workers can finish a job in 24 days. After 8 days 10 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

END OF QUESTION PAPER — **Total: 40 Marks**